

## EDUCATION

### University of Pittsburgh, PA, USA

Ph.D., Information Science Program, GPA: 3.986/4.0

Advisor: Dr. Peter Brusilovsky; Co-advisor: Dr. Morgan R. Frank

Dissertation Title: *Explainable Course Recommendation: Connecting College Education to Knowledge and Careers Through Skills*

Committee: Dr. Peter Brusilovsky, Dr. Morgan R. Frank, Dr. Daqing He, Dr. Zachary A. Pardos

### University of Information Technology, Ho Chi Minh City, Vietnam

M.S., Computer Science; Advisor: Dr. Nhon Do

### University of Information Technology, Ho Chi Minh City, Vietnam

B.S., Computer Science, Honors Program, rank: 2/101; Advisor: Dr. Nhon Do

## RESEARCH INTERESTS

Large Language Models, Agentic and Conversational AI  
Machine Learning and Deep Learning  
Educational Data Mining and Pattern Discovery  
Natural Language Processing (Knowledge Extraction and Representation)  
Personalized Ranking and Recommendation

## SKILLS

- Languages: Python, Java, R, Matlab, NodeJS, C++, C# .Net, SQL, HTML, CSS, JavaScript, Maple
- Deep Learning toolkits: TensorFlow, Pytorch, Keras
- Machine Learning toolkits: scipy, scikit-learn, statsmodels, CatBoost, SVM-light, R packages
- Toolkits: Claude Code, Cursor, Langchain, OpenAI, RAGAS, HuggingFace, numpy, matplotlib, t-SNE, NLTK, Spacy, Stanford NLP, D3.js

## INDUSTRY EXPERIENCE

### Zillow Group, Seattle, USA

Jan 2024 – Now

Applied Scientist, Agentic AI

- **Built (independently) the end-to-end evaluation framework for agentic AI systems from 0 → production adoption**, introducing **LLM-as-a-Judge**, ensemble scoring, multi-response evaluation, and **assertion-based evaluation** to address **non-determinism** and improve **measurement reliability**—powering evaluation of Zillow’s conversational AI Mode throughout the development lifecycle **through to launch**, and enabling benchmarking and incremental quality improvements across agent workflows.
- Led **comparative benchmarking of agentic systems vs. frontier models** (e.g., GPT-5, Claude), incorporating tool use and context-aware reasoning to identify capability gaps and guide modeling improvements.
- Applied **LLM-based modeling techniques** (classification, routing, triaging) to replace heuristic systems, improving performance on unstructured, high-ambiguity inputs within agent workflows.
- Developed reusable **agentic components and pipelines** (RAG, intent classification, safety classifiers, curriculum learning) forming the foundation of scalable, tool-using AI assistants.
- Designed and deployed a **goal-oriented conversational AI system**, enabling multi-turn task completion through structured planning, tool use, and state management.

**Technical skills:** Claude Code, Cursor, Agentic AI, LLMs, RAGAS, HuggingFace, Databricks, MLFlow, AWS, Langchain, Langgraph

### Zillow Group, Seattle, USA

Jan 2022 – Jan 2024

Applied Scientist, Search & Personalized Ranking

- Developed and deployed multiple **personalized ranking models** for home listings by leveraging large-scale user engagement signals, improving click-through and conversion metrics across Zillow Home Search.
- Developed and evaluated Zillow’s **first production-grade natural language search** capability for real estate discovery, enabling users to express complex housing preferences in free text and significantly improving search relevance.

**Technical skills:** Python, PySpark, PyTorch, AWS, HuggingFace, statsmodels, CatBoost, scikit-learn, KubeFlow

### Zillow Group, Seattle, USA

- Applied Scientist Intern  
May 2020 – Aug 2020
  - Leveraged **user engagement data** on Zillow websites to define **home demand** signals on the real estate market, and built machine learning models to improve **home resale prediction** values
  - **Technical skills:** Python, PySpark, AWS, statsmodels, CatBoost, scikit-learn
- Applied Scientist Intern  
Sep 2019 – May 2020
  - Implemented an Keyphrase Annotation system and built Logistic Regression and Bi-LSTM Deep Learning models to extract **keyphrases** from home listing descriptions for homes on Zillow websites
  - **Technical skills:** Python, PyTorch, AWS, NLTK, Spacy, Stanford NLP toolkits, Gensim

- Machine Learning Engineer Intern May 2019 – Aug 2019
  - Applied **deep learning** techniques and **regular expressions** to detect customers' personal identifiable information (**PII**) to protect their privacy in a *big data* setting
  - **Technical skills:** PyTorch, Python, AWS, NLTK, SQL, Airflow

**Lac Viet Computing Corporation, Vietnam**  
Software Engineer

Mar 2013 – Sep 2015

Built **mobile applications** on Windows Phone OS such as Lac Viet Dictionary, Lac Viet Reader, Words Usage Practice.

**Technical skills:** C# .Net, Windows Phone, Silverlight, SQLite

## RESEARCH EXPERIENCE

Member of “*Development and Implementation of a Health E-Librarian with Personalized Recommendations (HELPeR)*” project, funded by National Institutes of Health (NIH) U.S, led by Dr. Young Ji Lee, Dr. Peter Brusilovsky and Dr. Daqing He. Working on **extracting concepts** in medical text documents using **NLP** and **ML classification** and **deep sequential models**, and working on *personalized recommendation*.

**Technical skills:** Python, PyTorch, Spacy, Stanford POS tagger, scikit-learn, SVMlight, matplotlib, Jupyter notebook  
(01/2020 – 05/2020)

Collaborating with Dr. Zachary Pardos in “*Improving Articulation of Credit and Transfer Student Support at CUNY*” project. Working on **learning representation** and **semi-supervised/unsupervised machine translation** techniques to articulate courses for multiple institutes in New York city.

**Technical skills:** PyTorch, Keras, Python, matplotlib, t-SNE, D3 visualization library, Jupyter notebook  
(09/2019 – 01/2020)

Collaborated with Dr. Zachary Pardos in “*Deep Learning in Higher Education Big Data to Explore Latent Student Archetypes and Knowledge Profiles*” project. Working on **learning representation** and **supervised machine translation** techniques to find equivalent courses across colleges.

**Technical skills:** Keras, Python, matplotlib, t-SNE, D3 visualization library, Jupyter notebook  
(05/2018 – 08/2019)

Member of “*Open Corpus Personalized Learning*” project, funded by National Science Foundation (NFS), U.S, led by Dr. Peter Brusilovsky and Dr. Daqing He. Working on **extracting concepts** in textbooks using **NLP** and **ML classification models**, and working on **predicting student performance** in class.

**Technical skills:** Java, Python, HTML, Javascript, SQL, Stanford POS tagger, scikit-learn, SVMlight, matplotlib, R, R Markdown, Jupyter notebook  
(08/2016 – 09/2019)

Member of “*Researching and Developing the Sematic-based Solution for Learning Resources Management*” project, funded by VNU-HCM, worked on revising and improving the CK-ONTO model in as **document retrieval** system design.

**Technical skills:** HTML, Javascript, C# .Net, SQL  
(06/2014 – 06/2015)

## PUBLICATIONS

**Hung Chau**, Run Yu, Zachary Pardos, Peter Brusilovsky (2025)  
*Skill-based Explanations for Serendipitous Course Recommendation*  
arXiv:2508.19569

Jiaxin Xu, **Hung Chau**, Angela Burden (2025)  
*From Abstract to Actionable: Pairwise Shapley Values for Explainable AI*  
arXiv:2502.12525

**Hung Chau** (2024)  
Committee Members: Peter Brusilovsky, Morgan Frank, Zach Pardos, Daqing He  
*Explainable Course Recommendation: Connecting College Education to Knowledge and Careers Through Skills*  
University of Pittsburgh 2024

**Hung Chau**, Sarah H. Bana, Baptiste Bouvier, Morgan R. Frank (2023)  
*Connecting Higher Education to Workplace Activities and Earnings*  
PLOS One 2023

Mengdi Wang\*, **Hung Chau\***, Khushboo Thaker, Peter Brusilovsky, Daqing He (2021)  
*Knowledge Annotation for Intelligent Textbooks*  
Journal of Technology, Knowledge and learning (TKNL) 2021

Run Yu, Zach Pardos, **Hung Chau**, Peter Brusilovsky (2021)  
*Orienting Students to Course Recommendations Using Three Types of Explanation*  
Adjunct Proceedings of the 29th Conference on User Modeling, Adaptation & Personalization, June 21-25, 2021, Utrecht, Netherlands

**Hung Chau**, Saeid Balaneshin, Kai Liu, Ondrej Linda (2020)  
*Understanding the Tradeoff between Cost and Quality of Expert Annotations for Keyphrase Extraction*  
The 14th Linguistic Annotation Workshop (COLING 2020), Dec 12, 2020, Barcelona, Spain

**Hung Chau**, Igor Labutov, Khushboo Thaker, Daqing He, Peter Brusilovsky (2020)  
*Automatic Concept Extraction for Domain and Student Modeling in Adaptive Textbooks*  
International Journal of Artificial Intelligence in Education (JAIED) 2020

Behnam Rahdari, Peter Brusilovsky, Khushboo Thaker, **Hung Chau** (2020)

*CovEx: An Exploratory Search System for COVID-19 Scientific Literature*

The 3rd International workshop on Epidemiology meets Data Mining and Knowledge discovery, Aug 24, 2020, San Diego, CA, USA

Zachary Pardos, **Hung Chau**, Marshall Zhao (2019)

*Data-Assistive Course-to-Course Articulation Using Machine Translation*

Proceedings of the 6th ACM Conference on Learning @ Scale (L@S) June 24-25, 2019, Chicago, Illinois, USA [**Best paper award**]

**Hung Chau**, Jordan Barria-Pineda, Peter Brusilovsky (2018)

*Course-Adaptive Content Recommender for Course Authoring*

Proceedings of the 14th European Conference on Technology Enhanced Learning (EC-TEL), September 06-09, 2018, Leeds, UK

**Hung Chau**, Jordan Barria-Pineda, Peter Brusilovsky (2018)

*Course-Adaptive Content Recommender System for Instructors of Programming Courses*

Proceedings of the 19th International Conference on Artificial Intelligence in Education (AIED), June 27-30, 2018, London, UK

**Hung Chau**, Jordan Barria-Pineda, Peter Brusilovsky (2017)

*Content Wizard: Concept-Based Recommender System for Instructors of Programming Courses*

Adjunct Proceedings of the 25th Conference on User Modeling, Adaptation & Personalization, July 09-12, 2017, Bratislava, Slovakia

**Hung Chau\***, Ang Li\* and Yu-Ru Lin (2017)

*Predicting Students Performance Based on Their Reading Behaviors*

Online Proceedings of SBP-BRiMS 2017, July 05-8, 2017, Washington DC, USA

## HONORS AND AWARDS

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- Best Paper Award at ACM Conference on Learning @ Scale 2019
- 03/2016: VEF (an independent U.S. Federal Government agency created by the U.S. Congress) 2015 Fellow
- 11/2009: "SINH VIEN 3 TOT" certificate for excellent performance in Academic Accomplishment, Studiousness, and Physical
- 2009: The SMBC Global Foundation Scholarship Award; Institute of International Education
- 2007 – 2009: Certificates of Satisfactory Progress from the rector of UIT by having excellent academic accomplishment and studiousness

## CERTIFICATES

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- LearnLab: Simon Initiative Summer School at CMU. Advised by Dr. Zach Pardos, University of California at Berkeley

## OTHERS

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- Github: <https://github.com/HungChau>
- HuggingFace: <https://huggingface.co/HungChau>
- Google Scholar: <https://scholar.google.com/citations?user=eqdfdmIAAAAJ&hl=en>